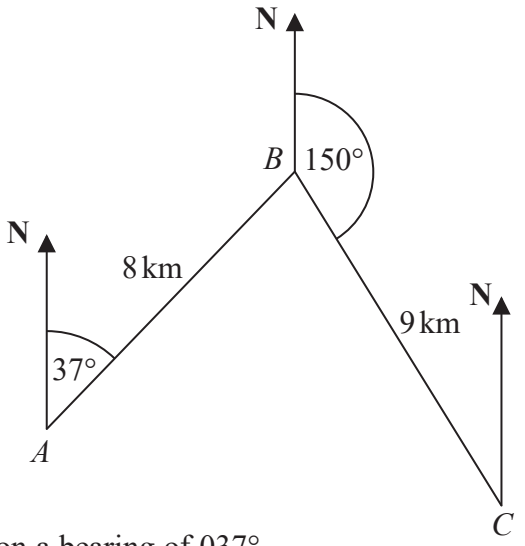
23 The diagram shows the positions of three towns, Acton (A), Barston (B) and Chorlton (C).



Barston is 8 km from Acton on a bearing of 037°
Chorlton is 9 km from Barston on a bearing of 150°

Find the bearing of Chorlton from Acton.
Give your answer correct to 1 decimal place.
You must show all your working.

BA bearing = 037° + 180° = 217°.

Angle ABC = 217° − 150° = 67°.

1 mark
Angle ABC = 67°.

$AC^2 = 8^2 + 9^2 - 2(8)(9)\cos 67°$.

1 mark
Cosine rule for AC.

AC = 9.4199... km.

sin(angle BAC)/9 = sin 67°/AC.

1 mark
Sine rule for BAC.

Angle BAC = 61.57...°.

Bearing of C from A = 37° + 61.57...°

1 mark
Add BAC to bearing 037°.

= 098.6° (1 d.p.).

1 mark
Bearing 098.6°.

098.6 °

(Total for Question 23 is 5 marks)